HDOC - Day 13

C Programming Activity | Menu-Driven Program using switch-case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits and odd digits. Example: Number = 12045 -> Even digits = 3, Odd digits = 2.
- Find the sum of even digits and the sum of odd digits separately. Example: Number = 12345 -> Sum of even digits = 6, Sum of odd digits = 9.

Algorithm

1. Start.

2. Input a positive integer num.

3. If num <= 0, display "Please enter a positive integer" and stop.

4. Display the menu: (1) Count even and odd digits, (2) Sum even and odd digits.

5. Input the user's choice.

6. Set temp = num.

7. Use switch(choice).

8. For Case 1, initialize even_count = 0 and odd_count = 0. Repeat while temp > 0: digit = temp % 10; if digit % 2 == 0 increment even_count, otherwise increment odd_count; then temp = temp / 10. Display both counts.

9. For Case 2, initialize sum_even = 0 and sum_odd = 0. Repeat while temp > 0: digit = temp % 10; if digit % 2 == 0 add digit to sum_even, otherwise add digit to sum_odd; then temp = temp / 10. Display both sums.

10. For any other choice, display "Invalid choice".

11. Stop.

C Program

```
#include <stdio.h>

int main() {
    int num, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
```

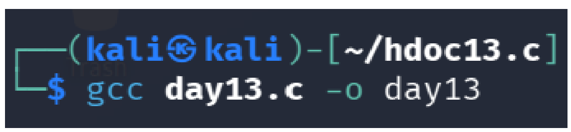
```c
    scanf("%d", &num);

    if (num <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Count even and odd digits\n");
    printf("2. Sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = num;

    switch (choice) {
        case 1:
            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;
                temp = temp / 10;
            }
            printf("Even digits = %d\n", even_count);
            printf("Odd digits = %d\n", odd_count);
            break;

        case 2:
            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;
                temp = temp / 10;
            }
            printf("Sum of even digits = %d\n", sum_even);
            printf("Sum of odd digits = %d\n", sum_odd);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Test Case Table

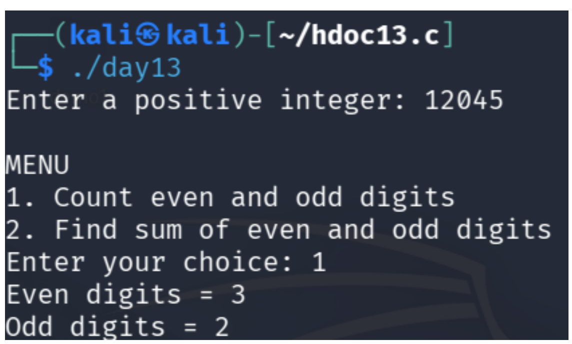
Test Case	Input Number	Choice	Expected Output	Purpose
1	12045	1	Even digits = 3 Odd digits = 2	Normal case: count digits
2	12345	2	Sum of even digits = 6 Sum of odd digits = 9	Normal case: sum digits
3	24680	1	Even digits = 5 Odd digits = 0	All digits even
4	13579	2	Sum of even digits = 0 Sum of odd digits = 25	All digits odd
5	8	1	Even digits = 1 Odd digits = 0	Single-digit input
6	0	1	Please enter a positive integer.	Invalid number
7	-123	2	Please enter a positive integer.	Negative input
8	12345	3	Invalid choice.	Invalid menu choice

Compilation and Execution

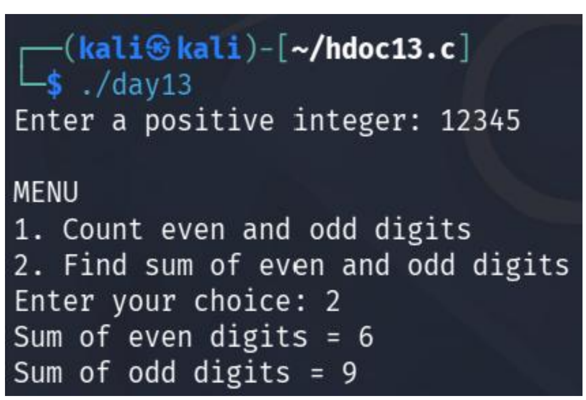
Compilation



Execution - Test Case 1



Execution - Test Case 2



Execution - Test Case 3

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: 24680

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 5
Odd digits = 0
```

Execution - Test Case 4

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: 13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum of even digits = 0
Sum of odd digits = 25
```

Execution - Test Case 5

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: 8

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 1
Odd digits = 0
```

Execution - Test Case 6

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: 0
Please enter a positive integer.
```

Execution - Test Case 7

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: -123
Please enter a positive integer.
```

Execution - Test Case 8

```
┌──(kali㉿kali)-[~/hdoc13.c]
└─$ ./day13
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 3
Invalid choice.
```

Result

The menu-driven C program was written, compiled, executed, and tested successfully for the prepared test cases.